

Activity 01: Data Cleaning & Visuals

SISMID Lab

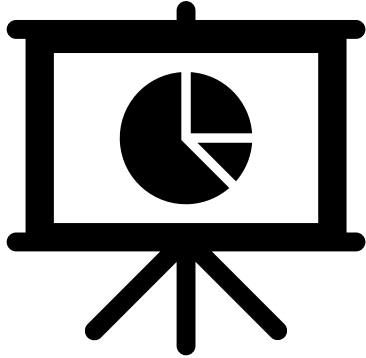
July 22 through July 24, 2026

A Human-AI Teaming Approach to Infectious Disease Modeling

Lecture Overview

Topic	Slides
1. General Housekeeping	3
2. Lab Timeline	4
3. Task Overviews	5
4. Goal Output	6
5. Data Cleaning: The Rules	7–11
6. Activity: Data Cleaning	12–13
7. Reference Files: agents.md & R Scripts	14–16
8. Results: Epicurve & Dataset	17–18

General Housekeeping

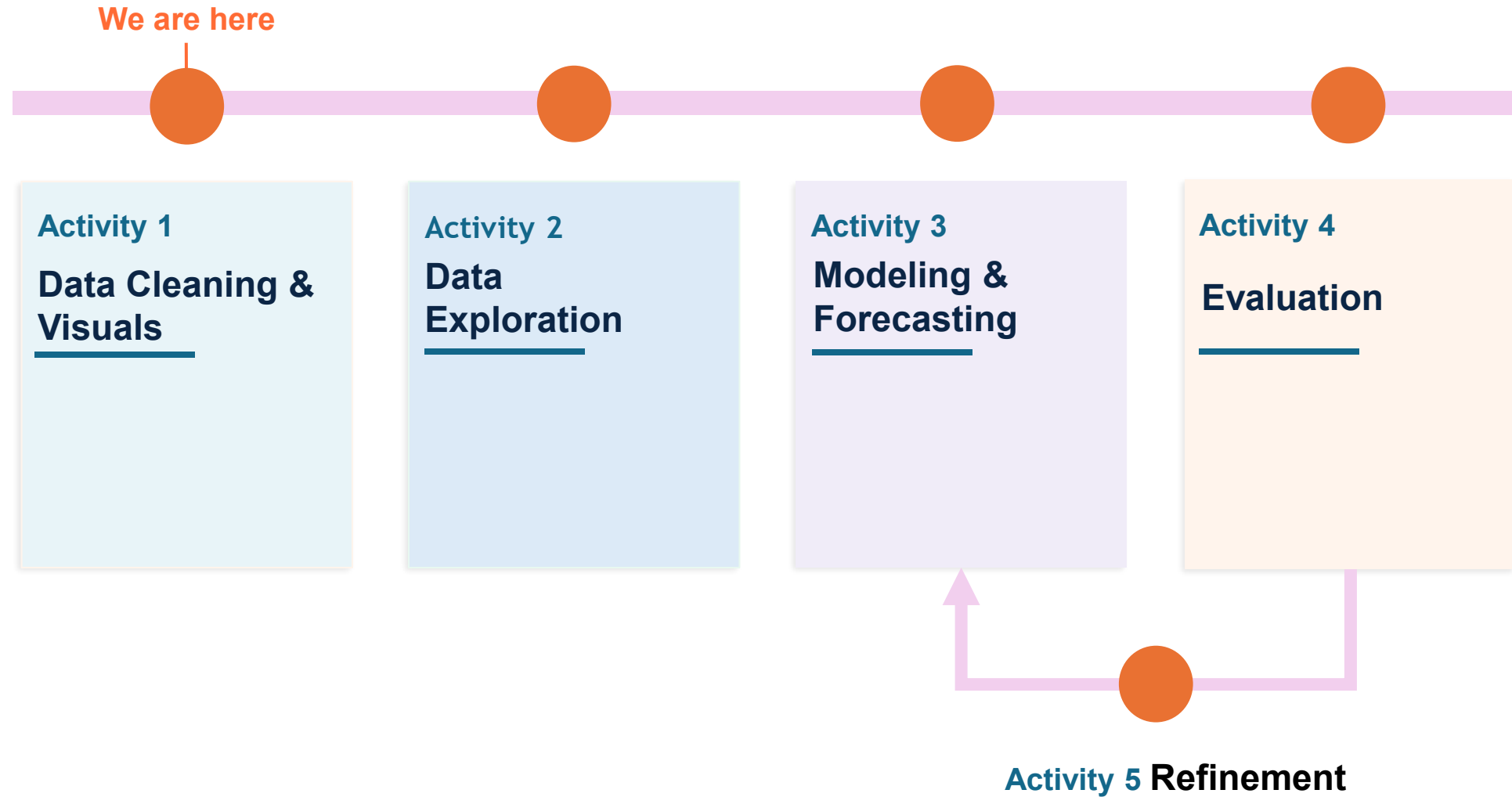


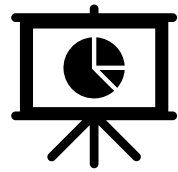
Any slide with this icon contains the **overview** and **expected outputs** for the included tasks. All outputs listed on these slides should be uploaded to the course GitHub repository for review.



Any slide with this icon indicates that it is **time to complete a hands-on activity**. Carefully follow the instructions provided, complete the required task, and verify your results.

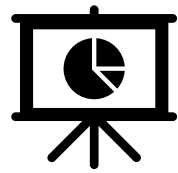
Lab Timeline





Task Overviews

Task	Description
Task 1: Confirm your data	Open the project folder and check it contains the <i>Weekly Hospital Respiratory Data (HRD) Metrics by Jurisdiction.csv</i> dataset.
Task 2: Review the rules	Open <i>rules.md</i> and confirm the <i>Data Cleaning Rules</i> match those covered in the lecture.
Task 3: Create <i>agents.md</i>	Prompt Copilot (or your AI of choice) to turn the rules document into an <i>agents.md</i> file.
Task 4: Build the cleaning script	Use <i>agents.md</i> with Copilot (or your AI of choice) to generate the <i>scripts/01_cleaning.R</i> script.
Task 5: Run & verify output	Run the script and confirm the cleaned CSV has three columns — week, location, value — and produces the epicurve.



Goal Output

From the activity (see previous slide), you should produce your own **cleaned dataset** and an **epicurve** visualization. Guided by your data-cleaning rules, an AI agent turns the raw Weekly HRD Metrics file into a tidy three-column dataset and the epidemic curve that summarizes it.

Activity Output	Description
cleaned_flu_admissions.csv	A tidy dataset with three columns — week, location, and value — output by the cleaning script.
Epicurve	An epidemic curve plotting the cleaned weekly admissions, summarizing the data visually.

Data Cleaning: The Rules

These rules are provided in the **rules.md** file. Your agent will follow them to generate **01_cleaning.R**.

1 Load the data

Read the *Weekly Hospital Respiratory Data (HRD) Metrics by Jurisdiction.csv* from the `data/` folder using `readr`.

2 Filter to US only

Column: 'Geographic aggregation' → keep rows where value is "USA".

3 Select target column

Keep only the 'Total Influenza Admissions' column. Drop all other columns, including those for RSV and COVID-19.

4 Reshape to 3 cols

Rename and restructure the data to: week, location ("US"), value.

5 Format dates

Convert week ending date to R Date object. Sort ascending.

6 Save output

Write *cleaned_flu_admissions.csv* and the epicurve to the appropriate subfolder in `output/`.

Reminder: Rules Document vs AI Agent

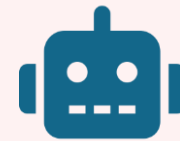


`rules.md`

Written by YOU

- Human-readable specifications
- Describes WHAT the pipeline should do
- You edit this to improve results

AI
generates



`agents.md`

Generated by AI

- AI-readable agent instructions
- Describes HOW to execute the pipeline
- The intermediary between you and code

At the end of the day, **you are still the expert** making the decisions.

rules.md Starting Place

```
cleaned_flu_admissions.csv U rules.md U X
rules.md > abc # Data Cleaning Rules > abc ## 8. Required validation checks
1 # Data Cleaning Rules
2
3 These rules define how the agent should generate `01_cleaning.R`. Follow each
4 step in order. The goal is to turn the raw NHSN HRD influenza file into a tidy,
5 three-column dataset ready for downstream modeling and produce an epicurve
6 figure.
7
8 ## 1. Load the data
9
10 Read the NHSN HRD CSV from the `data/` folder using **readr** (read_csv()).
11 To avoid parsing warnings from unrelated fields, import only required columns:
12
13 - `Week Ending Date`
14 - `Geographic aggregation`
15 - Influenza admissions column (see Rule 3)
16
17 Import these columns as character first, then parse/convert explicitly.
18
19 ## 2. Filter to US only
20
21 Use the `Geographic aggregation` column. Keep only the rows where the value is
22 `USA`.
23
24 ## 3. Select the target column
25
26 Use influenza admissions from one of these allowed column names:
27
28 - `Total.Influenza.Admissions`
29 - `Total Influenza Admissions`
30
```

We **do not** need to make any changes to the *rules.md* file for this activity. During future activities, we will be adding our own rules to this document and re-generating the *agents.md* file. For now, this is included as an example of how to structure the rules document as we build it up.

Prompting Your Agent

This activity will require two prompts: one to build the agent and one to generate code.

Step 1: Generate the agent

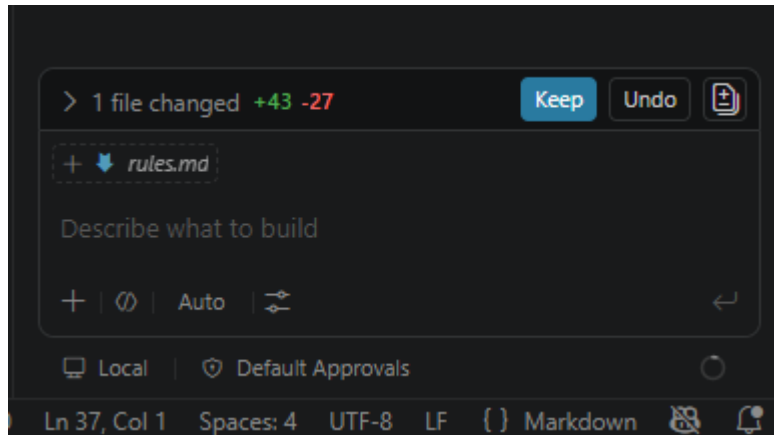
```
Using rules.md, generate an agent file and save as agents.md
```

Step 2: Generate the R script

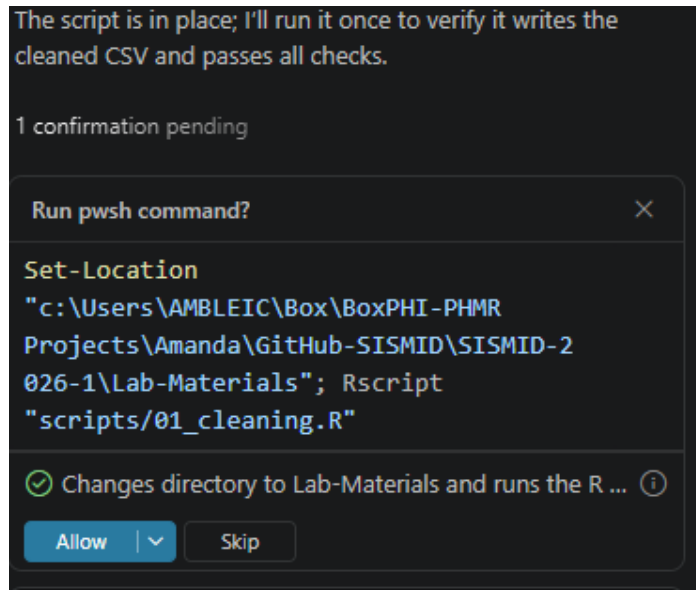
```
Using agents.md, generate a Data Cleaning script and save as scripts/01_cleaning.R
```

The rules doc is what YOU control. The agent is what the AI builds from it. The scripts are what the agent produces.

Some Tips



Each time we make a change to a file via the integrated AI, the changes made from file version to file version are tracked. Each time you make a change, you will want to see what the AI did, and if happy with the change, click **Keep**. This will change the file in your working directory to match that with the changes made by the AI.



The AI may try to run the code directly within the chat window. This can be buggy at times, so we suggest skipping this suggestion and running locally (via the editor window). If you decide to run via the chat window, you may be given a series of prompts until R is found on your local machine. To skip and run locally, click **Skip**.



Now it's your turn!

Together, we will work through Tasks 1 through 5 to create the **agents.md** file & data cleaning script, as well as prepare the data and get the epicurve.

- 1 Open the project folder and double-check that it contains *data/Weekly Hospital Respiratory Data (HRD) Metrics by Jurisdiction.csv*
- 2 Open the **rules.md** file and read *Data Cleaning Rules*. Make sure all rules match those specified on the earlier slide. If you have any questions, please let us know.
- 3 Prompt Copilot (or your AI of choice) to generate the **agents.md** file from the rules document (**Hint**: copy the Step 1 prompt from Slide 10).
- 4 Use the **agents.md** file and Copilot (or your AI of choice) to generate **scripts/01_cleaning.R** (**Hint**: copy the Step 2 prompt from Slide 10).
- 5 Run the script, check the output CSV has 3 columns: *week*, *location*, *value* and you have the epicurve of the data.

Expected Process & Results

agents.md File

CHAT

← GENERATE AGENT FILE FROM RULES.MD

Using rules.md, generate an agent file and save it as agents.md

I'll read the current `rules.md` and then create agents.md by translating those rules into a clear agent instruction file.

Reviewed rules.md

I've extracted the cleaning requirements from `rules.md`, and next I'll write a new agents.md that encodes them as executable agent instructions for producing the cleaning script.

Reviewed Lab-Materials and created agents.md

Created `agents.md` from the rules in `rules.md`.

It now contains:

1. A clear objective for generating `@1_cleaning.R`

2. Ordered workflow steps matching your rules

3. Input/output paths

4. Required package list

5. A strict output contract (exact 3-column schema)

6. Acceptance checks to verify the result

If you want, I can also generate the actual scripts/`01_cleaning.R` file from this agent spec next.

↻

📄

👍

🗨

> 1 file changed +43 -27

Keep

Undo

📄

+ `rules.md`

Describe what to build

+ | 🔍 | Auto | ⚙

📄 Local

🛡 Default Approvals

Ln 37, Col 1 Spaces: 4 UTF-8 LF () Markdown

```
agents.md > abc # Agent: NHSN HRD Influenza Data Cleaning > abc ## Purpose

# Cleaning Agent Instructions

These instructions define how the agent should generate `01_cleaning.R`. Follow
each step in order. The goal is to turn the raw NHSN HRD influenza file into a
tidy, three-column dataset ready for downstream modeling.

## 1. Load the data

Read the NHSN HRD CSV from the `data/` folder using `readr::read_csv()`.

1 # Agent Instructions: Build `01_cleaning.R`
2
3 ## Objective
4 Generate an R script named `01_cleaning.R` that cleans NHSN HRD influenza data into a tidy,
three-column dataset for downstream modeling and creates a US epicurve figure.
5

## 2. Filter to US only

Use the `Geographic aggregation` column. Keep only rows where the value is
`"USA"`.

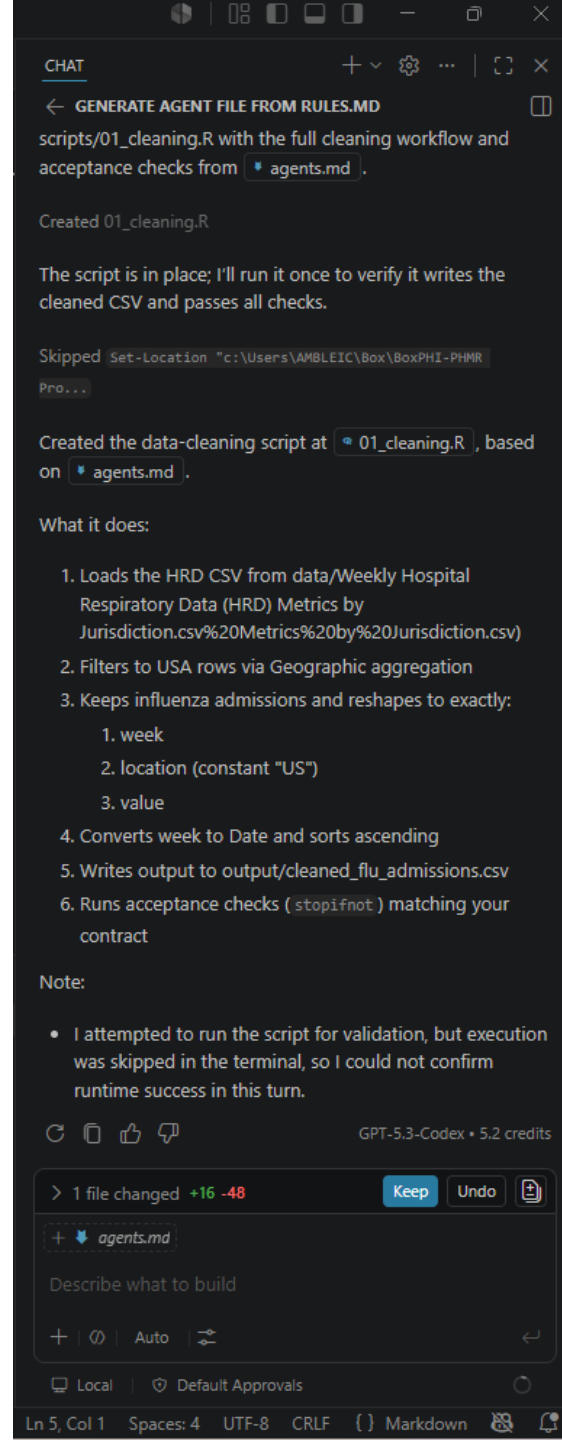
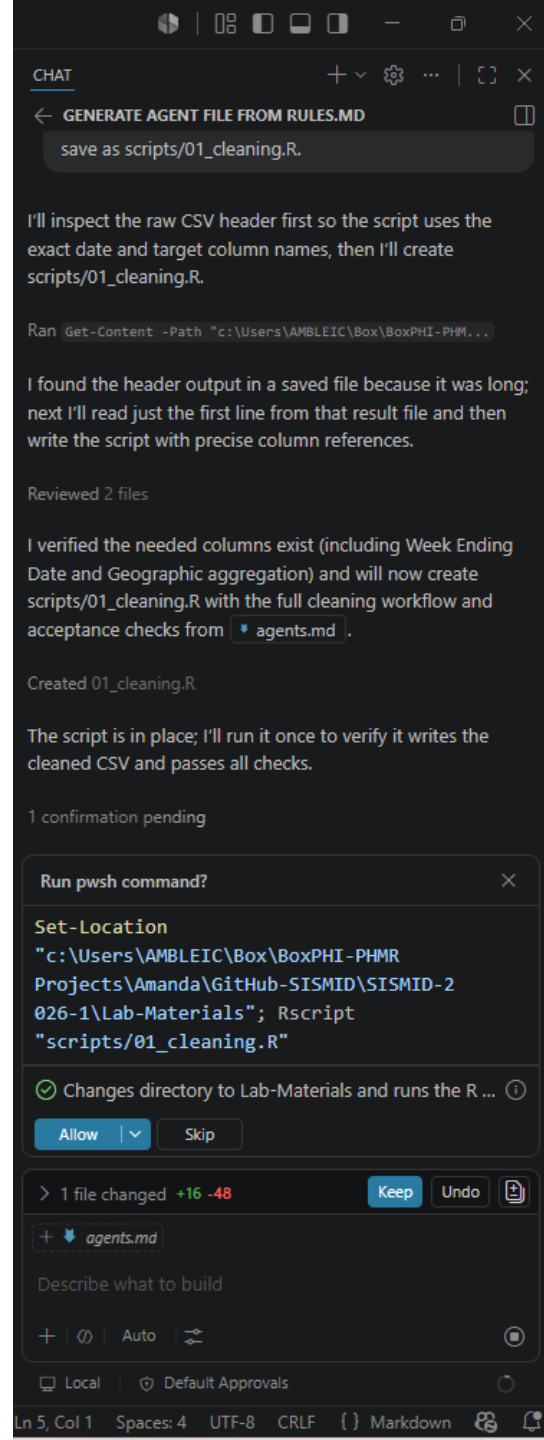
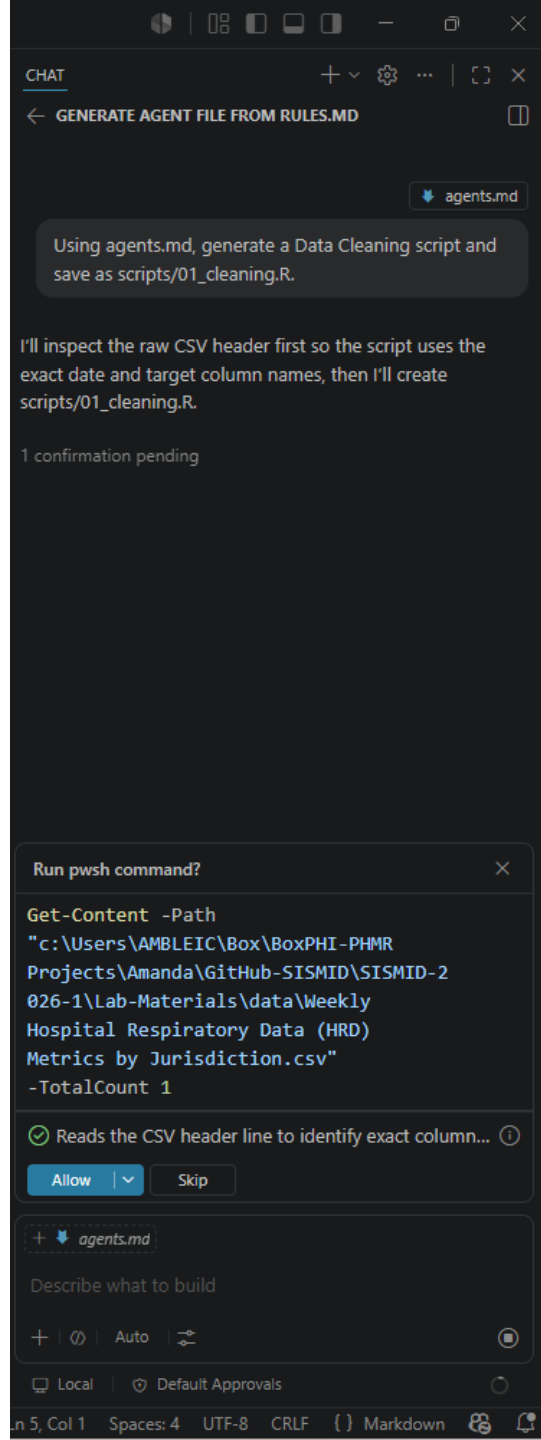
## 3. Select the target column

Keep only the `Total.Influenza.Admissions` column. Drop all COVID and RSV
columns.

## 4. Reshape to three columns

Rename and restructure the data to exactly three columns:

- `week`
```



R Script

When generating the *01_cleaning.R* script, the AI may prompt you multiple times at many steps from your initial message to the script.

For example, in the initial picture, it is asking permission to look at the data. In the second picture, it is asking permission to run the script (Skipped for this activity; see earlier slide).

The key is to read **WHY** you are being asked a question before clicking **Allow** or **Keep**.

01_cleaning.R File

scripts > 01_cleaning.R > cleaned

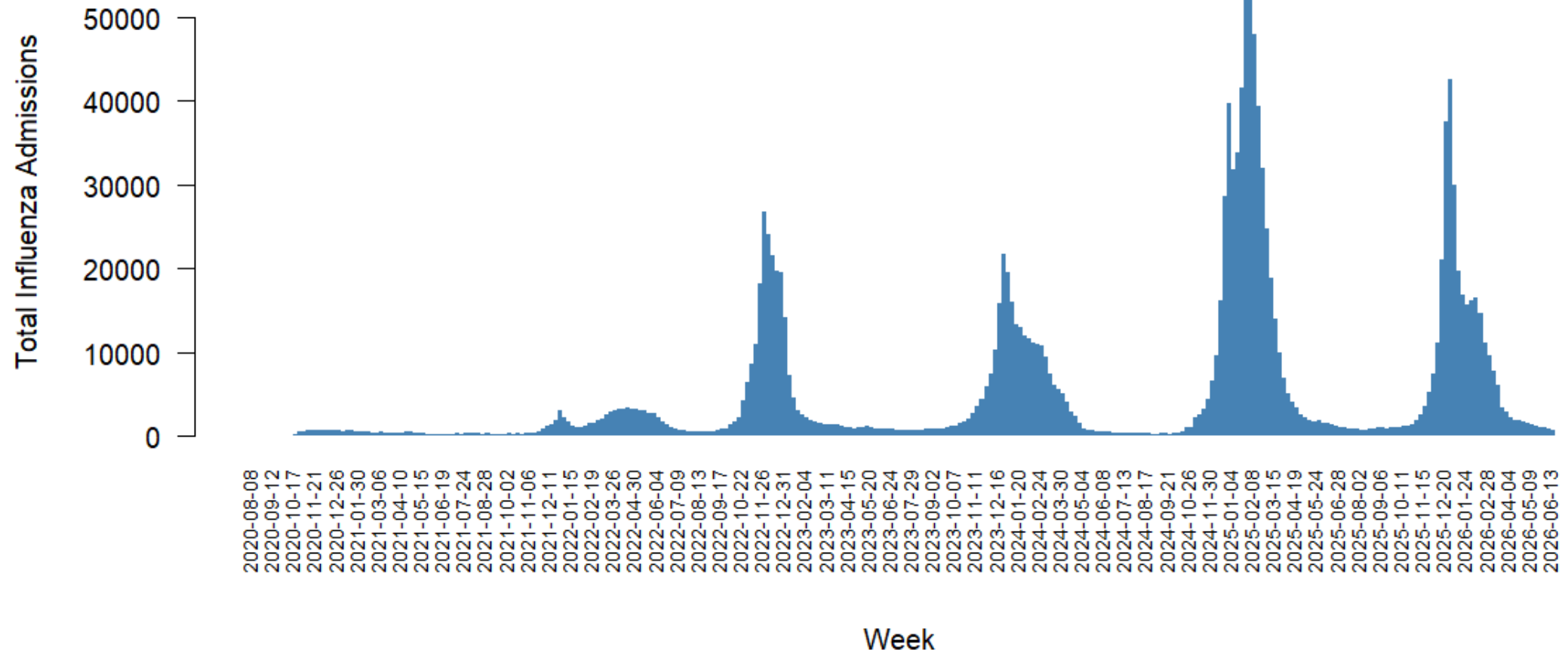
```
1 library(readr)
2 library(dplyr)
3
  input_file <- file.path("data", "Weekly Hospital Respiratory Data (HRD) Metrics by
  output_file <- file.path("output", "cleaned_flu_admissions.csv")

  dir.create("output", showWarnings = FALSE, recursive = TRUE)

  cleaned_data <- read_csv(input_file, show_col_types = FALSE, na = c("", "NA")) |>
    filter(`Geographic aggregation` == "USA") |>
4 input_path <- "data/Weekly Hospital Respiratory Data (HRD) Metrics by Jurisdiction.csv"
5 output_path <- "output/cleaned_flu_admissions.csv"
6 figure_path <- "output/figures/epicurve_us_flu_admissions.png"
7
8 raw <- read_csv(
9   input_path,
10   col_select = c(
11     `Week Ending Date`,
12     `Geographic aggregation`,
13     any_of(c("Total.Influenza.Admissions", "Total Influenza Admissions"))
14   ),
15   col_types = cols(.default = col_character()),
16   show_col_types = FALSE
17 )
18
19 # Support both dotted and spaced versions of the influenza admissions column name.
20 influenza_col <- c("Total.Influenza.Admissions", "Total Influenza Admissions")
21 influenza_col <- influenza_col[influenza_col %in% names(raw)][1]
22
23 if (is.na(influenza_col)) {
```


Resulting Epicurve

US Weekly Influenza Admissions Epicurve



Resulting Dataset: *cleaned_flu_admissions.csv*

[illegible]